

LAEGEREN, Switzerland

WMO: 066690

Lat: 47.4819N

Lon: 8.3972E

Elev: 2766

StdP: 13.29

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 13.0	(c) 16.8	(d) 8.9	(e) 9.6	(f) 18.5	(g) 12.7	(h) 11.6	(i) 21.3	(j) 30.5	(k) 38.7	(l) 27.7	(m) 38.9	(n) 7.0	(o) 30	(p) 0.563

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	80.4	64.0	77.1	62.6	74.2	61.7	65.7	76.3	64.3	74.0	62.9	72.0	3.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.9	91.9	69.9	60.6	87.5	68.3	59.3	83.6	66.8	32.1	76.7	30.9	74.2	29.9	72.2	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.5	20.9	18.4	10.4	84.3	5.0	3.6	6.9	86.9	4.0	89.0	1.2	91.0	-2.5	93.6
			9.9	68.6	4.8	1.8	6.4	69.9	3.6	71.0	0.9	72.0	-2.7	73.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.6	30.5	32.1	38.8	45.6	52.9	59.6	62.7	62.2	54.7	47.7	38.4	32.5
	DBStd	13.70	7.48	8.46	8.06	8.31	7.43	7.68	6.80	6.87	6.62	7.39	7.56	7.37
	HDD50	2763	604	501	356	180	54	9	1	2	28	131	354	543
	HDD65	6882	1069	921	814	581	379	198	126	138	314	535	799	1008
	CDD50	1507	0	0	7	49	145	296	395	381	168	61	5	0
	CDD65	152	0	0	0	0	6	35	55	52	4	0	0	0
	CDH74	688	0	0	0	1	13	157	266	248	3	0	0	0
	CDH80	104	0	0	0	0	0	20	42	42	0	0	0	0
(14) Wind	WSAvg	6.6	7.6	7.5	7.4	7.0	6.1	5.5	5.6	5.5	6.0	6.4	7.0	7.4
(15) Precipitation	PrecAvg	44.5	2.8	2.7	2.8	3.3	4.8	4.9	5.0	4.7	3.4	3.4	3.5	3.4
	PrecMax	57.3	6.7	6.5	9.2	6.6	9.1	8.1	8.9	8.7	6.9	6.0	7.6	7.3
	PrecMin	33.3	0.5	0.8	0.9	0.4	1.2	2.0	1.7	0.7	1.5	0.4	0.2	0.0
	PrecStd	6.1	1.7	1.5	1.8	1.7	2.0	1.7	1.9	1.8	1.4	1.5	1.9	1.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.0	52.1	60.7	69.6	75.7	82.9	84.5	84.5	73.5	66.2	57.9	51.2
		MCWB	42.9	42.3	48.1	52.8	59.2	65.2	65.0	63.5	61.3	55.3	48.1	43.0
	2%	DB	45.5	48.8	56.6	65.3	71.7	78.5	80.2	80.1	70.7	62.8	54.5	46.8
		MCWB	40.8	40.8	45.6	50.7	57.5	63.0	63.9	64.1	60.1	54.7	47.2	42.2
	5%	DB	43.4	45.8	53.5	61.6	68.1	75.0	76.8	76.3	67.6	60.2	51.3	44.4
		MCWB	39.3	39.6	44.3	49.5	55.5	61.8	62.7	62.8	59.0	53.8	46.1	40.9
	10%	DB	40.6	43.1	50.2	58.2	64.7	71.6	73.5	73.2	64.7	57.8	48.6	42.2
		MCWB	37.7	38.5	42.9	47.8	54.0	60.5	61.6	62.1	57.7	52.9	44.9	39.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.5	44.6	49.3	54.3	61.5	67.7	67.7	67.3	63.6	58.2	51.0	46.0
		MCDB	46.6	47.7	57.0	65.9	71.4	79.9	79.9	77.1	70.6	62.7	54.0	47.5
	2%	WB	42.5	42.2	47.3	52.0	59.1	64.9	65.5	65.5	61.6	56.4	48.8	43.6
		MCDB	44.1	46.5	53.8	62.3	68.6	74.8	76.4	75.9	68.1	60.4	52.0	45.6
	5%	WB	40.1	40.6	45.6	50.3	57.1	63.0	63.9	64.1	59.9	55.0	47.1	42.0
		MCDB	42.2	44.7	51.2	59.5	65.6	72.2	73.8	73.8	65.8	58.9	50.2	44.0
	10%	WB	38.0	38.9	43.7	48.6	55.1	61.4	62.6	62.7	58.3	53.4	45.0	40.0
		MCDB	40.1	42.4	49.1	56.9	63.1	69.9	71.7	71.6	63.7	56.9	48.2	41.9
(35) Mean Daily Temperature Range	5% DB	MDBR	6.5	7.4	9.9	12.0	12.2	12.2	12.3	11.3	10.0	8.2	6.8	6.6
		MCDBR	8.8	11.1	13.6	15.9	15.7	15.7	16.0	15.0	13.2	10.7	10.6	9.0
		MCWBR	6.2	6.5	7.0	7.7	7.9	7.6	7.1	6.7	6.6	5.9	5.6	5.9
	5% WB	MCDBR	7.9	9.7	12.3	14.8	14.5	14.5	14.7	14.0	12.0	9.7	9.3	7.8
		MCWBR	6.7	6.7	7.3	7.6	7.9	7.6	7.2	6.9	6.5	6.0	6.2	6.2
(40) Clear-Sky Solar Irradiance	taub		0.281	0.304	0.336	0.364	0.376	0.380	0.381	0.374	0.352	0.332	0.302	0.277
	taud		2.480	2.420	2.383	2.331	2.332	2.355	2.341	2.375	2.423	2.470	2.507	2.482
	Ebn at Noon		259	274	281	282	282	280	278	275	272	260	248	246
	Edh at Noon		20	26	32	37	38	38	38	35	30	25	20	18
(44) All-Sky Solar Radiation	RadAvg		360	625	1009	1373	1587	1825	1760	1536	1151	721	398	295
	RadStd		40	87	137	197	159	170	161	157	117	77	48	47

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	+1.13	+0.77	N/A	N/A	+135	+52	(46)
(47)	Regional (116 neighbors)	N/A	N/A	N/A	+1.26	+0.76	+0.61	N/A	N/A	+101	+27	(47)

Nomenclature: See separate page